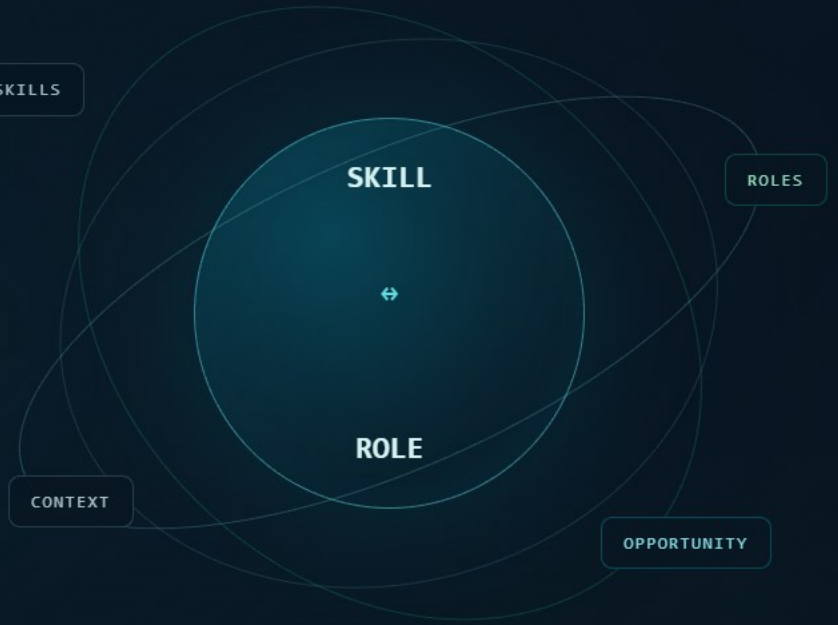
SDC HACKATHON 2026 · INTELLIGENT WORKFORCE NAVIGATION

SkillVerse AI

From existing skills to future-ready careers.

UN SDG 4 — Quality Education

Domain 1 Track



TEAM
I Love You 3000

INSTITUTION
Universiti Teknologi Malaysia (UTM)

MEMBERS
GUI GUANYU & NG JAA WEI

Assisted by Replit | SDC Hackathon 2026

Skills move faster than career systems.

Three forces leave learners behind. SkillVerse turns them into a clear route forward.

Problem Statement

Where current career systems fall short

Skills Go Outdated Fast
Tech and AI change daily, leaving textbooks and courses behind.

Job Roles Are Too Fixed
Standard career databases take years to add new, in-demand roles.

No Clear Learning Path
People learn blindly without knowing their exact skill gaps.

Solution

Five-step navigation pipeline

01
Profile
- Upload PDF or image resume.
- AI extracts skills safely.
- Strips personal info.

→

02
Live Trends
- Scans real job & research data.
- Finds fast-rising skills.
- Real-time market demand.

→

03
Role Match
- Math-backed role scoring.
- Finds exact skill gaps.
- No AI guesswork or bias.

→

04
Roadmap
- Multi-phase interactive path.
- Step-by-step learning.
- Targets what you lack.

→

05
Progress
- Quizzes & gamified XP.
- Auto-adapts to new trends.
- Exportable PDF reports.

Architected for Zero Demographic Bias & In-Memory Data Privacy.

Telemetry → Matching → Action.

Future-role matching.

Deterministic by design.

THE CORE HYBRID MATCHING FORMULA

Match Score = 0.60 × Macro + 0.40 × Micro

Transparent mathematical evaluation eliminating black-box generative hallucinations.

60%

MACRO · SEMANTIC ALIGNMENT

Independent Inputs · Dense Vectors

INPUT A · CANDIDATE

Candidate Profile

Skills, Titles, Context

INPUT B · VERIFIED BENCHMARK

ESCO Role Definitions

Title, Industry, Scope

BAAI/bge-m3

1024-dim · each input embedded independently.

Cosine similarity → Macro Vector Proximity (60% Weight)

40%

MICRO · COMPETENCY FIT

3-Tier Granular Cascade

1 — EXACT MATCH

Strict Token & Exact Substring

2 — JACCARD

Token Jaccard ≥ 60%

3 — DENSE SEMANTIC

Dense Cosine Semantic Vector

$cos(Vec_U, Vec_R) \geq 0.58$

03 / DYNAMIC KNOWLEDGE ENGINE

Living Data: Evolving Beyond Static Taxonomies

CONTINUOUS TELEMETRY LOOP

Traditional registries take years to refresh. SkillVerse AI keeps taxonomies evergreen through daily automated updates from live telemetry.

01 SIGNAL SOURCES

Telemetry Feeds

A **ArXiv cs.AI**
Research abstracts

R **RemoteOK**
Live job demand

stream

02 FRONTIER DETECTION

Velocity Tracker



Computes real-time market velocity and detects **unindexed frontier domains.**

03 DUAL-DATABASE DAILY SYNC

Dual-Database Synchronization

esco_roles starts from verified European ESCO seed data. Both tables receive automated daily updates.

Daily Cron Pipeline | ESCO Seed + Daily Upsert | Live Trend Table

TABLE A · esco_roles
European ESCO base seed + daily upsert of newly emerging frontier roles & skill vectors.

TABLE B · trend_signals
Daily refresh of skill velocities V_s and labor demand heat.

DAILY TAXONOMY UPDATE

Anchor: Official ESCO base seed.
Daily Evolution: Dynamic injection of emerging roles and trend velocity signals directly into Supabase pgvector.

Data Foundations: European ESCO Taxonomy · ArXiv cs.AI · RemoteOK

All content is for reference purposes only; final decision rests with the user.

A comprehensive system built for real execution.

EMPOWERING INDIVIDUAL LEARNERS & ENTERPRISES

01

Personalized Roadmap

Interactive React Flow node graph across Foundational, Core, and Frontier phases.

React Flow | 3-Phase Path

02

Deficit Heatmap & PDF

Analyze gaps from permitted peers/teams via Canvas; export directly to styled PDF reports.

Peer Permission | PDF Export

03

Gamified XP & Quizzes

Milestone self-assessments, continuous XP accumulation, and level mastery unlocking.

AI Quizzes | XP Progression

04

User-Driven Replanning

Recalculates the full reskilling trajectory on demand against newly fetched market trends.

User Autonomy | Trend-Aware

05

Trilingual Accessibility

Native instantaneous switching between English, Bahasa Melayu, and Chinese (中文).

EN / BM / ZH | next-intl

06

Mobile & Dark/Light Mode

Responsive mobile layout with persistent Dark/Light UI theme customisation.

Responsive | Theme Toggle

LIVE DEMO

INTERACTIVE PROTOTYPE WALKTHROUGH